

HAT-P-32b

R-1114

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_201214 · created 2026-07-30T21:04:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459197.7611 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.038
Transit depth (Rp/Rs)^2	6.8e-07 +/- 0.00063 [%]
Orbital Inclination (inc)	87.12 +/- 1.76
Airmass coefficient 1 (a1)	0.59 +/- 0.11
Airmass coefficient 2 (a2)	0.368 +/- 0.09
Scatter in the residuals of the lightcurve fit is	16.61 %
Best Comparison Star	None
Optimal Aperture	4.07
Optimal Annulus	11.9
Transit Duration (day)	0.1127 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.038 vs 0.1489 (deviation 1.53 σ)
Duration fit vs published	0.1127 d vs 0.1304 d (deviation 1.44 σ)
Mid-time O-C	-5.8 min
Depth SNR	0.0
Residual scatter	16.61 %

Light curve

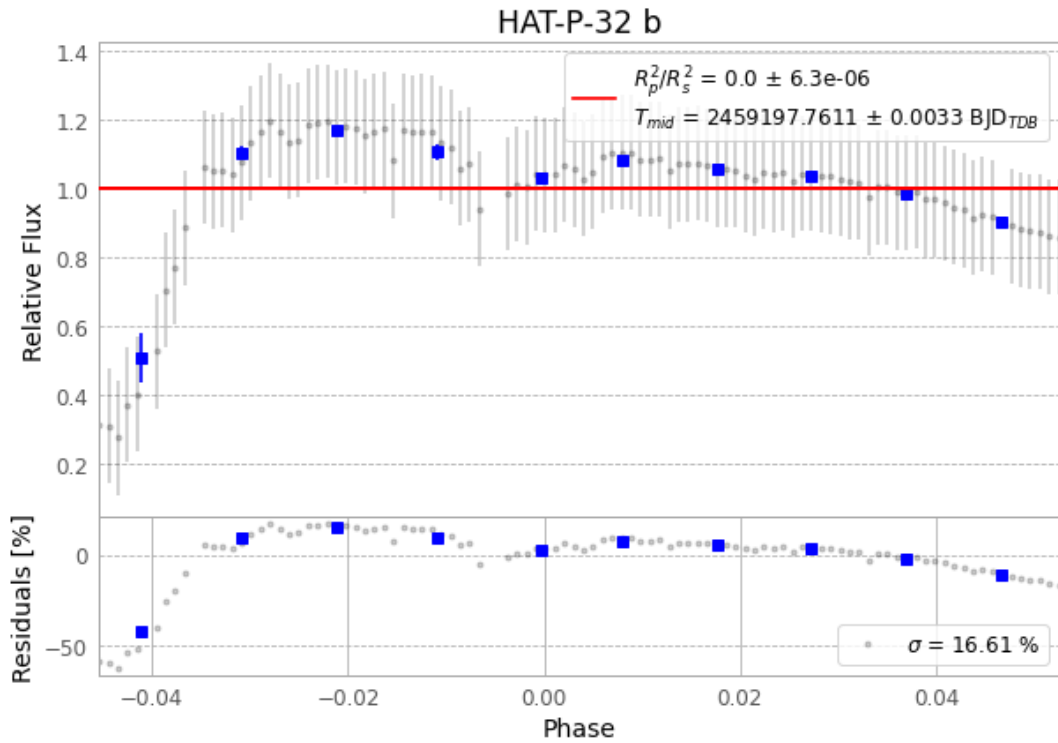


Figure 1 — FinalLightCurve_HAT-P-32 b_2020-12-13.png (SHA-256 969ae9da8ea46f4a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [515, 362], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a319891f075dd719...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), HATP-32201214034512.FITS → HATP-32201214085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-201214.log (88973424b745...), FinalLightCurve_HAT-P-32 b_2020-12-13.png (969ae9da8ea4...), FinalLightCurve_HAT-P-32 b_2020-12-13.csv (6ed6358a883d...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:04Z.